

Foundation Models in Agriculture: A Comprehensive Review

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Abstract: This paper explores the transformative potential of Foundation Models (FMs) in agriculture, driven by the need for efficient and intelligent decision support systems in the face of growing global population and climate change. It begins by outlining the development history of FMs, including general FM training processes, application trends and challenges, before focusing on Agricultural Foundation Models (AFMs). The paper examines the diversity and applications of AFMs in areas like crop classification, pest detection, and crop image segmentation, and delves into specific use cases such as agricultural knowledge question-answering, image and video analysis, decision support, and robotics. Furthermore, it discusses the challenges faced by AFMs, including data acquisition, training efficiency, data shift, and practical application challenges. Finally, the paper discusses future development directions for AFMs, emphasizing multimodal applications, integrating AFMs across the agricultural and food sectors, and intelligent decision-making systems, ultimately aiming to promote the digitalization and intelligent transformation of agriculture.

Keywords: foundation models; smart agriculture; language foundation models; vision foundation models; multimodal foundation models; agriculture foundation models



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1. Introduction

With the rapid development of Artificial Intelligence (AI) technology, research and application of Foundation Models (FMs) have gradually become a focal point across various fields. In agriculture, traditional AI applications mainly rely on specific Machine Learning (ML) and Deep Learning (DL) technologies [1], such as crop classification and pest detection. However, in recent years, the introduction of FM technology has brought new transformative opportunities to agriculture [2]. These models, which are pre-trained on a large scale and fine-tuned for specific agricultural tasks, can handle multimodal data, including text, images, audio, and video, and have the ability for cross-domain transfer learning. The goal of FMs is to enhance the intelligence level of agricultural production, optimize resource management, improve crop yield and quality, and solve practical problems in agricultural production.

The motivation for writing this paper stems from the urgent need for efficient, intelligent decision-support systems in agriculture. With the increasing global population and the impact of climate change, traditional agriculture faces unprecedented challenges, such as resource wastage, low production efficiency, and environmental issues. Therefore,

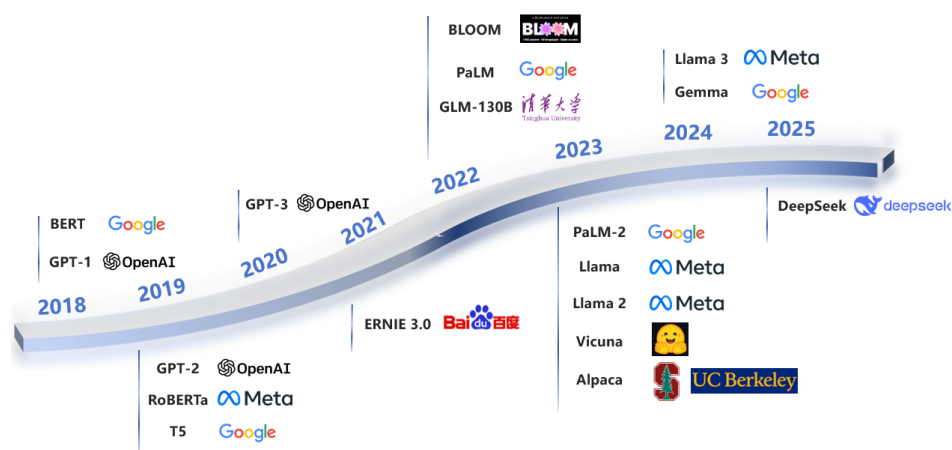
leveraging advanced FM technologies to improve the intelligent and precise management of agricultural production has significant practical value and broad application prospects. By exploring the application of FMs in agriculture, we can drive the sector toward more precise, efficient, and sustainable practices, addressing the current challenges in agriculture and advancing the digitalization and intelligent transformation of the industry.

We systematically collected 84 relevant publications published between 2019 and 2025, which provided an empirical foundation for this study and identified critical technological gaps in the field of agricultural large models.

This study makes four key contributions to advance AFMs:

- We systematically introduce the development of general-purpose foundation models in computer science, including their technical evolution and core architectures (Figure 1, Tables 1 and 2), providing essential background for non-computer science researchers to understand these transformative AI technologies.
- We comprehensively review existing AFMs and analyze their agricultural applications in knowledge Q&A, disease detection, and decision support (Table 3), offering practical insights for domain experts.
- We identify unique challenges in developing AFMs, including agricultural data heterogeneity, temporal shifts in field conditions, and deployment constraints for smallholder farms.
- We propose future directions emphasizing multimodal integration and intelligent decision systems to bridge AI innovation with agricultural needs.

By synthesizing insights from a large body of research, this work serves as a technical reference for interdisciplinary researchers and an implementation guide for agricultural practitioners.

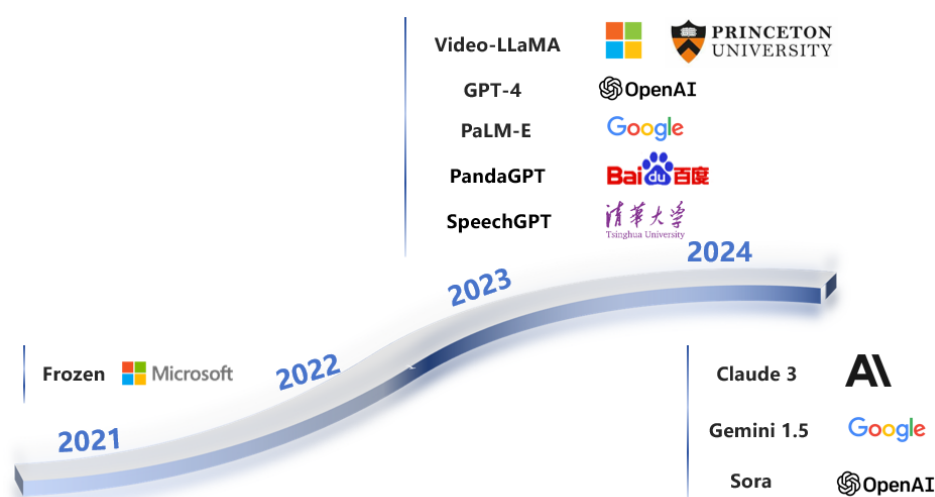


(a) NLP-based foundation models timeline

Figure 1. Cont.



(b) Visual-based foundation models timeline



(c) Multimodal-based foundation models timeline

Figure 1. Timeline of existing foundation models (FMs) in recent years, categorized by: (a) NLP-based models; (b) Visual-based models; and (c) Multimodal-based models. Timeline data was extracted from the original papers' release dates.

2. Overview of FM Development

In this section, we provide an overview of the technical background and development history of FMs, examining their historical development, key technological evolution, and significance. This section analyzes the core technologies of FMs, presents the classification of FMs, and offers a reference table that covers different types of FMs and their applications. This section further elaborates on the construction process of FMs, encompassing data processing, model architecture, training optimization, and evaluation. Additionally, it examines the applications of FMs and the challenges associated with them.

2.1. History of FMs

A FM is an artificial intelligence model trained on extensive data with large-scale parameters and complex computational structures [3], typically constructed from deep neural networks [4] and characterized by large-scale parameters, enabling adaptation to a wide range of downstream tasks [5], such as speech recognition, Natural Language Processing (NLP) [6–9], Computer Vision (CV) [10–13], and decision-making [14,15]. From a technological perspective, deep neural networks underpin FMs, with both technologies having evolved over decades.

The evolution of machine learning within AI has progressed through distinct phases, beginning with its emergence in the 1990s as a paradigm for automated model building from training data [3] to the recent dominance of DL. While early approaches relied on domain-specific feature engineering for complex tasks in NLP and CV [5], DL's multi-layered architectures facilitates automated feature extraction, thereby revolutionizing pattern recognition through large-scale datasets and enhanced computational resources [16]. This shift, exemplified by breakthroughs in image recognition [17,18], has achieved superhuman performance in controlled settings [19,20] but faces deployment challenges in practical applications due to interpretability and data quality concerns.

By the end of 2018, the field of NLP was on the verge of another seismic shift, ushering in the era of FMs [5]. Artificial intelligence technology has evolved from small-scale data to big data, from limited models to FMs, and from specialized to general-purpose systems, steadily entering the FM era. Currently, research on FMs is primarily focused on NLP, CV, and multimodal tasks [21]. FMs are classified according to their capacity into NLP-based FMs, vision-based FMs, and multimodal-based FMs. In Figure 1, we outline the evolutionary processes of these three types of FMs in detail.

Large Language Models (LLMs) represent advanced neural architectures characterized by their massive scale, typically comprising billions or more parameters. These extensive models demonstrate unique emergent capabilities that distinguish them from conventional Pre-trained Language Models (PLMs) [22]. Empirical evidence suggests a positive correlation between model scale and performance, with larger architectures exhibiting enhanced sample efficiency and superior task-solving capabilities [23]. Comparative analyses between scaled PLM implementations reveal significant behavioral differences, where expanded models demonstrate remarkable proficiency in addressing complex challenges [24]. A notable manifestation of these emergent properties is evident in GPT-3's [7] contextual learning capacity, which enables the generation of targeted outputs through text sequence completion without additional parameter optimization, a capability not present in its predecessor, GPT-2 [6].

Large Vision Models (LVMs) represent a significant advancement in CV research. Early approaches in visual pattern recognition primarily relied on basic feature extraction techniques, such as scale-invariant feature transformation and gradient orientation histograms, which were subject to substantial constraints. The landscape transformed dramatically in 2012 when AlexNet's [17] remarkable performance in the ImageNet competition sparked widespread adoption of Convolutional Neural Networks (CNNs). Subsequent architectural innovations, including VGGNet [25], GoogLeNet [26], and ResNet [27], significantly enhanced the capabilities in image analysis tasks. The proliferation of internet-scale image datasets facilitated the development of advanced detection frameworks, such as Faster R-CNN [28], YOLO [29], and Mask R-CNN [30]. More recently, the integration of Transformer architectures with Generative Adversarial Networks (GANs) has yielded groundbreaking models, such as Vision Transformer(ViT) [31] and DALL-E [32], showcasing exceptional performance in visual understanding and synthesis through self-attention mechanisms.

Multimodal Large Language Models (MLLMs) [33] have emerged as a pivotal advancement in AI research, addressing the limitations of single-modal systems. While LLMs excel in textual tasks and LVMs demonstrate proficiency in visual analysis, both exhibit limited capabilities when processing cross-modal data. MLLMs overcome these limitations by integrating diverse data modalities, including visual, textual, and auditory information, thereby enabling comprehensive multimodal understanding and processing. This technological breakthrough has significantly expanded AI's capabilities, facilitating sophisticated interpretation and manipulation of heterogeneous data types [33].

To provide a comprehensive overview of the development of FMs, we summarize the key models in NLP, vision, and multimodal domains in Tables 1 and 2, including their architectures, parameters, and applications.

Table 1. Overview of popular FMs (Part 1).

Model Type	Model Name	Model Creators	Release Year	# Architecture	# Parameters
NLP-based	GPT-1 [34]	OpenAI	2018	Decoder only	117M
	GPT-2 [6]	OpenAI	2019	Decoder only	117 M, 345 M, 762 M, 1.5 B
	GPT-3 [7]	OpenAI	2020	Decoder only	175 B
	GLM-130B [35]	Tsinghua University	2022	Encoder–decoder	130 B
	BERT [36]	Google	2018	Encoder only	340 M
	RoBERTa [37]	Meta	2019	Encoder only	340 M
	T5 [38]	Google	2019	Encoder–decoder	60 M, 220 M, 770 M, 3 B, 11 B
	Gemma [39]	Google	2024	Decoder only	2 B, 7 B
	PaLM [40]	Google	2022	Decoder only	8 B, 62 B, 540 B
	PaLM-2 [41]	Google	2023	Decoder only	340 B
	BLOOM [42]	BigScience	2022	Decoder only	3 B, 7.1 B, 176 B
	ERNIE 3.0 [43]	Baidu	2021	Encoder only	27 M, 75 M, 118 M
	Llama [9]	Meta	2023	Decoder only	7 B, 13 B, 33 B, 65 B
	Llama 2 [9]	Meta	2023	Decoder only	7 B, 13 B, 34 B, 70 B
	Llama 3 [44]	Meta	2024	Decoder only	8 B, 70 B, 400 B
	Vicuna [45]	Hugging Face	2023	Decoder only	7 B, 13 B
	Alpaca [46]	Stanford and UC Berkeley	2023	Decoder only	7 B
	DeepSeek LLM [47]	DeepSeek AI	2024	Decoder only	7 B, 67 B
	DeepSeek-V2 [48]	DeepSeek AI	2024	Decoder only	236 B
	DeepSeek-V3 [48]	DeepSeek AI	2024	Decoder only	671 B
	DeepSeek-R1 [49]	DeepSeek AI	2025	Decoder only	671 B
Vision-based	LLaVA [50]	UC Berkeley and Microsoft	2023	Encoder only	13 B
	BLIP-2 [51]	Alibaba	2023	Encoder only	12 B
	Flamingo [52]	Google	2022	Encoder only	3 B, 9 B, 80 B
	Florence [53]	Microsoft	2021	Encoder only	893 M
	Florence 2 [54]	Microsoft	2024	Encoder–decoder	0.2 B, 0.7 B
	Segment Anything Model (SAM) [12]	Meta	2023	Encoder–decoder	375 M, 1.25 G, 2.56 G
	UFO [55]	Baidu	2022	Encoder only	17 B
	INTERN [56]	SenseTime	2023	Encoder–decoder	20 B
	DALL-E [32]	OpenAI	2021	Encoder–decoder	12 B
	DeepSeek-VL [57]	DeepSeek AI	2024	Decoder only	1.3 B, 7 B
	DeepSeek-VL2 [58]	DeepSeek AI	2024	Decoder only	1.0 B, 2.8 B, 4.5 B
Multimodal	GPT-4 [59]	OpenAI	2023	Decoder only	1.8 T
	Sora [60]	OpenAI	2024	Encoder–decoder	-
	Claude 3 [61]	Anthropic	2024	Decoder only	20 B, 70 B, 2 T
	Video-LLaMA [62]	Princeton & Microsoft	2023	Encoder only	-
	Gemini 1.5 [63]	Google	2024	Decoder only	Nano 1.8 B/3.25 B
	PaLM-E [64]	Google	2023	Decoder only	562 B
	PandaGPT [65]	Baidu	2023	Decoder only	-
	SpeechGPT [66]	Tsinghua University	2023	Decoder only	13 B
	Frozen [67]	Microsoft	2021	Decoder only	7 B

Table 2. Overview of Popular FMs (Part 2).

Model Name	I->O Modality	Open Source	Key Applications	Pre-Train Data Scale	Agri-Trained	Agri-Applicable
GPT-1	Text -> Text	✓	Text generation, language modeling	-	×	×
GPT-2	Text -> Text	✓	Text generation, language modeling	-	×	×
GPT-3	Text -> Text	✓	Text generation, language modeling	5 G	×	×
GLM-130 B	Text -> Text	✓	Text generation, question answering	-	×	×
BERT	Text -> Text	×	Text generation, question answering, summarization	570 G	×	×
RoBERTa	Text -> Text	✓	Language understanding, text classification, NER	-	×	×
T5	Text -> Text	✓	Text generation, question answering, summarization	-	×	×
Gemma	Text -> Text	✓	Multi-language text generation, understanding	366 B	×	×
PaLM	Text -> Text	✓	Text generation, summarization, translation	750 G	×	×
PaLM-2	Text -> Text	✓	Text generation, summarization, creative writing	3 T, 6 T	×	×
BLOOM	Text -> Text	✓	Text classification, question answering	-	×	×
ERNIE 3.0	Text -> Text	×	Text understanding, knowledge-enhanced tasks	-	×	×
Llama	Text -> Text	✓	Cross-lingual tasks, text generation	780 B	×	×
Llama 2	Text -> Text	✓	Cross-lingual tasks, text generation	3.6 T	×	×
Llama 3	Text -> Text	✓	Text generation, question answering	2 T	×	✓
Vicuna	Text -> Text	✓	Chatbot, dialogue systems	-	×	×
Alpaca	Text -> Text	✓	Text generation, question answering	1.4 T	×	✓
DeepSeek LLM	Text -> Text	✓	Text generation, question answering	15 T	×	✓
DeepSeek-V2	Text -> Text	✓	Chatbot, dialogue systems	70 K samples	×	×
DeepSeek-V3	Text -> Text	✓	Instruction-following tasks	52 K samples	×	×
DeepSeek-R1	Text -> Text	✓	Text generation, dialogue systems	2T	×	×
LLaVA	Image + Text -> Text	✓	Vision and language understanding, image captioning	158 K	×	×
BLIP-2	Image + Text -> Text	✓	Visual Question Answering (VQA), image captioning	129 M	×	✓
Flamingo	Text + Image -> Text	✓	Reasoning, math, code generation	-	×	×
Florence	Image -> Image	×	Image recognition, visual understanding	-	×	×
Florence 2	Image -> Image	×	Image recognition, visual understanding	-	×	×
Segment Anything Model (SAM)	Image -> Segmentation masks	✓	Object segmentation, masking, image manipulation	1 B	×	✓
UFO	Image -> Image	×	Industrial visual inspection, object detection	-	×	×
INTERN	Image -> Image	×	Image recognition, visual understanding	-	×	×
DALL-E	Text -> Image	✓	Image generation from text descriptions	-	×	×
DeepDeepSeek-VL2Seek-VL	Image + Text -> Text	✓	Vision-language tasks	-	×	×
	Image + Text -> Text	✓	Vision-language tasks	-	×	×
GPT-4	Text + Image -> Text	×	Text generation, image understanding	1.8 T	×	✓
Sora	Text -> Video	×	Video generation	-	×	×
Claude 3	Text -> Text	×	Text generation, complex reasoning	-	×	×
Video-LLaMA	Video + Text -> Text	✓	Video understanding, video captioning	-	×	×
Gemini 1.5	Text + Image -> Text	×	Multimodal understanding	-	×	×
PaLM-E	Text + Image -> Text	✓	Robotics, multimodal reasoning	-	×	×
PandaGPT	Text + Image -> Text	✓	Multimodal understanding	-	×	×
SpeechGPT	Speech + Text -> Text	✓	Speech recognition and synthesis	60 K h	×	×
Frozen	Text + Image -> Text	✓	Multimodal learning	-	×	×

2.2. Process of Building FMs

2.2.1. Data Selection and Processing

Training data play a pivotal role in determining the performance of FMs. Training data can generally be classified into two main categories: general-purpose and domain-specific data. General-purpose data include a wide array of content, such as web pages, books, and conversational text, which is abundant and diverse but often contains noise. Domain-specific data, in contrast, refer to datasets tailored to particular fields, such as scientific papers, code, or multilingual texts, which are essential for refining the model's capabilities for targeted domains or tasks. Models like BLOOM [42] and PaLM [41] have leveraged such domain-specific datasets to improve performance across a range of languages and domains.

The processing of training data is a fundamental aspect of ensuring high-quality model performance [68]. For general-purpose data, particularly web data, the first step is to filter out low-quality content. This involves removing irrelevant information, such as spam, advertisements, or misleading data. Methods for filtering low-quality data can generally be categorized into classifier-based and heuristic-based approaches. Classifier-based methods, as employed in models such as GPT-3 [7] and PaLM [41], involve training a machine learning classifier to distinguish between high- and low-quality content. Heuristic-based methods, used by models like BLOOM [42], rely on predefined rules or patterns to identify and exclude undesirable data.

Given that web data are vast and heterogeneous, consisting of diverse text formats, topics, and writing styles, preprocessing must also address issues such as duplications and biases. The data must be cleaned and structured to ensure that the remaining content is both relevant and representative of the tasks the model is expected to handle. Another critical step is ensuring data privacy, which involves eliminating Personally Identifiable Information (PII) to comply with regulations such as GDPR [69].

Domain-specific datasets also require careful handling. Scientific literature, for instance, contains complex terminology that must be processed to enable the model to understand domain-specific nuances. Multilingual data present challenges related to diverse grammar rules, syntax, and language-specific structures, necessitating specific preprocessing techniques such as tokenization or translation normalization [70–72]. For code generation, it is essential to clean and format code snippets to ensure they can be properly interpreted by the model [73,74].

2.2.2. FMs Architectures

FMs are generally built upon three dominant architectural paradigms: encoder-only, decoder-only, and encoder–decoder models, most of which rely on the Transformer framework as the foundational building block [75].

(1) Encoder Only: Encoder-only architectures, such as BERT [36], are designed for tasks requiring a deep understanding of input sequences, such as named entity recognition, sentence classification, and extractive question answering. These models typically rely on a bidirectional self-attention mechanism, enabling the attention layers to access information across the entire input sequence. Pretraining objectives, such as Masked Language Modeling (MLM), involve corrupting parts of the input sequence and tasking the model with reconstructing it, thereby enhancing the model’s understanding of contextual relationships.

(2) Decoder Only: Decoder-only architectures, exemplified by GPT models [7], are autoregressive and optimized for text generation tasks. In these models, the attention layers can only access preceding tokens in the sequence, enabling them to predict the next token in a stepwise manner. The decoder-only paradigm is particularly suited for tasks such as story generation, conversational AI, and completion-based language modeling.

(3) Encoder–Decoder: Encoder–decoder architectures, such as T5 [38] and BART [76], integrate the strengths of both encoders and decoders. The encoder processes the input sequence in its entirety, while the decoder generates output sequences based on the encoded representation. These models are ideal for sequence-to-sequence tasks, such as machine translation, summarization, and generative question answering. Pretraining objectives often include span corruption, where random spans of text are masked and the model is tasked with reconstructing them.

2.2.3. Training and Optimization

The training of FMs involves addressing challenges related to computational efficiency, model scalability, and task-specific performance. The training process typically consists of two main stages: pre-training on vast amounts of data and fine-tuning to adapt the model to specific tasks. To address these challenges, a range of optimization techniques have been developed to enhance the efficiency and effectiveness of large-scale model training.

(1) Pre-training of FMs: During pre-training, FMs are exposed to vast corpora of unlabeled text, typically employing self-supervised learning techniques. Two common approaches to this process are autoregressive language modeling [77], where the model predicts the next token in a sequence, and masked language modeling [78], where specific tokens are masked, and the model learns to predict them based on the surrounding context. Recently, techniques such as Mixture of Experts (MoE) have emerged, where a sparse set of expert networks is dynamically selected during training to reduce computational overhead, enabling the scaling of both model size and dataset size without a proportional increase in computing costs [79].

(2) Fine-tuning for Task-Specific Adaptation: Fine-tuning [80] is essential for adapting pre-trained models to specific tasks. It can be broadly categorized into two approaches:

- **Instruction Tuning:** Instruction tuning [81] focuses on enhancing the model's ability to follow specific instructions across a variety of tasks. This process can be further categorized into full-parameter fine-tuning [82] and Parameter-Efficient Fine-Tuning (PEFT) [83]. Full fine-tuning updates all the parameters in the model to optimize performance for a specific task, but its high computational cost and resource requirements limit its practicality for many use cases. In contrast, PEFT methods achieve task-specific adaptation by updating a small subset of parameters, thereby significantly reducing computational overhead. A notable example of PEFT is LoRA (Low-Rank Adaptation) [84], which applies low-rank matrix approximations to weight updates, further reducing the number of trainable parameters and enhancing scalability.
- **Alignment Fine-Tuning:** Alignment fine-tuning [24] is designed to refine the model's behavior to align with human values and expectations. This approach often employs reinforcement learning techniques that incorporate feedback signals to guide the model's adjustments. A widely used method is Reinforcement Learning with Human Feedback (RLHF) [85], where human-provided feedback, such as preference rankings or annotations, serves as a reward signal for training the model. RLHF has been instrumental in improving the alignment of FMs with user preferences, ensuring that the generated outputs are more relevant and aligned with human expectations. A complementary approach, Reinforcement Learning with AI Feedback (RLAIF) [86], replaces human feedback with AI-generated signals, reducing reliance on human labor while maintaining training efficiency. This approach is particularly valuable for scaling alignment processes in FMs, where collecting human feedback at scale may be impractical.

(3) Optimization Techniques for Enhanced Performance: To further enhance the performance of FMs, Retrieval-Augmented Generation (RAG) [87] has emerged as an effective optimization technique. One of the primary limitations of pre-trained models is their inability to access up-to-date or domain-specific knowledge. RAG addresses this limitation by dynamically retrieving relevant information from external sources, such as search engines or knowledge graphs, and integrating it into the model's generation process. By integrating retrieval-based information, RAG enhances the relevance of responses, making it a valuable technique for a wide range of applications.

2.2.4. Evaluation Metrics

Evaluating FMs involves using various metrics to assess their performance across different tasks. The following outlines the key methods commonly used for evaluation.

(1) Task-Specific Metrics: Task-specific metrics are efficient and cost-effective, providing a rapid means of evaluating FM performance on defined tasks. Common metrics include ROUGE (Recall-Oriented Understudy for Gisting Evaluation) [88] for text summarization and text generation, which measures recall, precision, and F1 score, and BLEU (Bilingual Evaluation Understudy) [89] for machine translation. These metrics primarily focus on text quality and linguistic accuracy. However, their scope is limited to specific tasks and may not capture nuances such as style, cultural context, or the subtleties of natural language, which are essential for more complex or generalized applications of FMs.

(2) Research Benchmarks: Standardized research benchmarks, such as MMLU (Massive Multitask Language Understanding) [90], GLUE (General Language Understanding Evaluation) [91], and SuperGLUE [92], enable a broad and consistent evaluation of model performance across a variety of tasks and datasets. These benchmarks are useful for quickly assessing foundational model (FM) capabilities, as they cover a wide range of topics and problem domains. However, these benchmarks are not without limitations. Issues such as data contamination (i.e., overlap between benchmark datasets and model training data) [93]

and the possibility of “benchmark gaming” (where models are fine-tuned specifically to perform well on these tests) [94] can lead to unreliable assessments. Additionally, these benchmarks are often focused on specific evaluation dimensions, which may not fully capture a model’s versatility or real-world performance across varied tasks.

(3) Model Self-Evaluation: Model self-evaluation enables a pre-trained model to assess the performance of other models by calculating specific metrics, such as perplexity, diversity, and consistency [7]. This method is fast and straightforward, providing an internal mechanism for evaluating generated outputs or model performance. While it is effective for certain applications, such as verifying the accuracy of RAG systems [95], it is not without limitations. For instance, self-evaluation can be resource-intensive and sensitive to the model’s selection criteria and the prompts provided. Moreover, it struggles with tasks that require reasoning or advanced problem-solving, such as step-by-step mathematical reasoning [96].

(4) Human Evaluation: Human evaluation remains the most reliable method for assessing the quality of FMs, particularly in ensuring that their outputs align with human values and expectations. Human evaluators can rank models or evaluate outputs based on subjective criteria, such as coherence, relevance, and overall quality. Common approaches include crowdsourced evaluations [97], where multiple humans evaluate the same model output, and expert evaluations [97], which involve skilled human annotators providing in-depth analysis. While highly reliable, human evaluations are resource-intensive, time-consuming, and expensive, particularly when high-level expertise is required. These evaluations are invaluable for more subjective assessments but may lack efficiency for broader, general-purpose model benchmarking.

2.3. Applications and Challenges of FMs

With the continuous advancement of artificial intelligence, FMs are permeating various industries at an unprecedented speed, demonstrating revolutionary potential and significant application value. However, alongside the numerous innovative opportunities they offer, FMs also face substantial challenges in practical applications, including high computational resource consumption, data privacy and security issues, and limited model interpretability.

2.3.1. Applications of FMs

In recent years, the rapid development of FM technology has showcased its transformative capabilities due to its exceptional semantic understanding, multimodal processing, and logical reasoning skills. These models are profoundly reshaping various vertical domains, such as information retrieval, biotechnology, and autonomous driving. Specifically, in the domain of information retrieval, FMs can capture the true intent behind user queries through deep semantic understanding, thereby enhancing query efficiency, as demonstrated by the work of Mohammad Kachuee et al. [98], who employed query rewriting and expansion to clarify user needs. In biotechnology, FMs have become indispensable tools in scientific research, driving a shift from traditional experiment-driven approaches toward intelligent, data-driven methodologies, as exemplified by DeepMind’s AlphaFold2 [99], which significantly accelerates protein structure prediction and shortens research cycles. In the realm of autonomous driving, while conventional systems rely on multiple independent algorithmic modules, FMs integrate perception, decision-making, and control through deep learning to boost overall system intelligence. For instance, Wang et al. [100] introduced the DriveMLM framework, which achieves closed-loop autonomous driving by standardizing decision states and employing multimodal modeling. Furthermore, FMs are finding applications in smart cities, film production, intelligent education, and robotics [101]. In

summary, as FM technology continues to break traditional industry boundaries, it is poised to fundamentally transform production methods and daily lives, delivering unprecedented efficiency and possibilities.

2.3.2. Challenges of FMs

Challenge 1: Computational Resource and Training Duration

The primary challenge of FMs lies in their high computational resource demands during both training and inference [102]. Training these models requires immense computational power, heavily dependent on high-performance GPUs or TPUs, and imposes strict requirements on the scale and performance of computing infrastructure. Moreover, the enormous number of parameters and the need to store intermediate computation results significantly increase memory requirements, presenting notable challenges for hardware configurations. The high computational cost restricts the widespread application of FMs. It also affects their efficient deployment and sustainable operation. Additionally, the iterative parameter optimization process can span several weeks or even months, demanding substantial computational resources, time, and meticulous debugging to gradually achieve the desired performance targets. Consequently, enhancing computational efficiency and reducing resource consumption through algorithm optimization, hardware acceleration, and model compression has become a critical research focus in artificial intelligence.

Challenge 2: Data Privacy and Security

FMs typically require massive datasets for optimal performance, often containing sensitive user information [103]. Inadequate protection mechanisms can lead to data breaches and severe security risks. Furthermore, during both training and inference, data may be susceptible to adversarial attacks or malicious tampering, which can compromise the model's accuracy and robustness. Such issues not only threaten user privacy but also undermine the model's credibility and its effectiveness in practical applications, thereby posing significant challenges to the broad deployment and long-term development of AI systems.

Challenge 3: Model Interpretability and Transparency

Given that FMs often comprise billions of parameters, their complex decision-making processes and prediction outputs are difficult to interpret directly [24]. This lack of interpretability poses significant challenges, particularly in fields requiring high transparency and trust, such as medical diagnosis and financial risk management, where interpretability is fundamental for user confidence, regulatory compliance, and practical deployment. The opaque nature of these models may lead to user skepticism regarding their outputs, thereby limiting their application in critical tasks. Moreover, insufficient interpretability hinders error analysis and performance optimization, complicating the identification and correction of potential biases. Although both academia and industry have intensified research into enhancing the interpretability of FMs, achieving a balance between maintaining high performance and improving transparency remains a pressing challenge.

3. FMs in Agriculture

This section investigates the development, diversity, and applications of FMs within the agricultural domain. Figure 2 delineates the construction process of AFMs. We analyze various FMs utilized for tasks such as text classification, disease detection, and image segmentation. These models employ state-of-the-art architectures, including Transformer-based models and multimodal systems that integrate both text and image data. Furthermore, we explore optimization strategies and pre-training techniques tailored for specific agricultural applications. This section offers a comprehensive overview of key models and their contributions to agricultural research and practical applications. These advancements

underscore the growing diversity in model architectures and their adaptability to various agricultural challenges, which will be further explored in the following subsections.

The implementation of AFMs must address the fundamental dichotomy between smallholder and industrial-scale farming systems. Current research reveals that most AFM architectures follow distinct technological pathways for these operational scales due to divergent infrastructure requirements and economic realities. Smallholder systems, representing over 80% of global farms, require specialized models emphasizing three critical characteristics: offline functionality for connectivity-challenged environments, minimal hardware dependencies, and extreme cost efficiency. These constraints have driven innovations like WhatsApp-based advisory platforms and TinyML implementations capable of operating on low-power edge devices, which achieve 70–80% accuracy at radically reduced deployment costs [104,105]. In contrast, industrial agricultural operations employ high-accuracy (>90%) cloud-based AFMs that process terabytes of multispectral drone and IoT sensor data through sophisticated transformer architectures. This approach requires substantial infrastructure investments in 5G networks and computing resources. This technological bifurcation reflects underlying socioeconomic realities—where smallholder systems prioritize accessibility and resilience, industrial operations optimize for precision at scale through capital-intensive solutions [106].

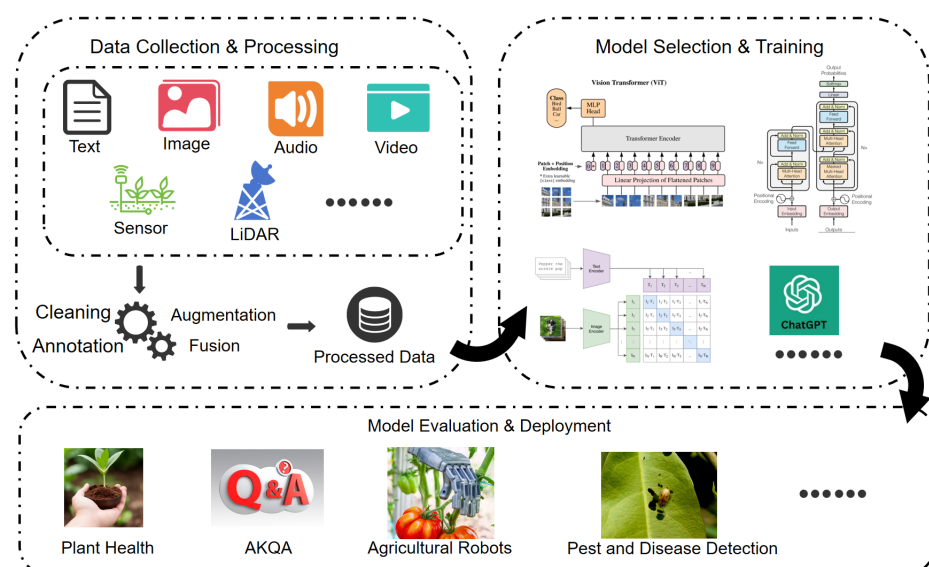


Figure 2. Development process of AFMs, divided into three main stages: data collection and processing, model selection and training, and model evaluation and deployment.

Table 3 summarizes some of the current AFMs. The emergence of AFMs began in 2022, with one of the earliest models being AgriBERT [107], which utilizes the BERT architecture and is primarily used for tasks such as text classification and generation on agricultural journals and USDA datasets. AgriBERT was pre-trained from scratch using a standard Masked Language Modeling (MLM) objective, where 15% of tokens in input sentences were randomly masked, and the model learned to predict them based on contextual information. The training corpus consisted of 46,446 food and agriculture-related journal articles (311 million tokens, 2.39 million words) rigorously cleaned to remove URLs, emails, and non-ASCII characters, supplemented by WikiText-103 and Penn Treebank datasets for linguistic diversity. As technological advancements occurred, the complexity of model architectures and tasks increased, leading to the introduction of new architectures such as ViT and RoBERTa, which offer stronger support for agricultural tasks involving both images and text.

The validation and processing of agricultural data begins with collecting multi-source texts (such as journal articles and extension materials) and converting them into standardized formats. This is followed by data cleaning to remove noise including garbled characters, non-UTF-8 characters, and the standardization of punctuation and capitalization. When constructing question-answering datasets, documents are filtered according to paragraph-length criteria to divide into training, validation, and test sets. Annotation tools are then employed to generate QA pairs where answers must precisely match source text segments without paraphrasing while ensuring diverse question types. Quality control is implemented through phased supervision by domain experts, including collaborative annotation with regular sampling checks during the annotation phase and consistency validation during evaluation. The inter-annotator agreement is quantified using both Cohen's Kappa and F1 scores, with the latter (reaching 0.86 in AgXQA1.1) proving more reliable for scenarios with multiple potential answers, thereby ensuring high data quality and task suitability [108].

Table 3. Overview of agricultural models and tasks.

Model Name	Year	Architecture	Dataset	Pre-training Method	Fine-tuning Method	Input → Output
AgriBERT ([107])	2022	BERT	Agricultural journals, USDA datasets	MLM trained from scratch	Adding external knowledge, fine-tuning	Text → classification labels, generated text
ITLMLP ([109])	2023	ViT-S/16 or ViT-B/16, CLIP	Cucumber, Apple, PlantVillage datasets	Image-text multimodal pre-training	Transfer pre-training, fine-tuning classification head	Image, text, labels → disease category labels
AgRoBERTa ([108])	2024	RoBERTa	Agricultural promotion corpus, AgXQA dataset	MLM pre-trained on AEC 1.1	Traditional fine-tuning, LoRA fine-tuning	Text → answer text
ChatAgri ([110])	2023	GPT3.5 (Transformer architecture)	Amazon Food, PestObserver, Agri-News, etc.	Pre-trained on ChatGPT results	Multiple prompt strategies and answer alignment strategies for fine-tuning	Text → classification labels
PLLaMa ([111])	2024	Based on LLaMa-2 (Transformer architecture)	Plant science academic papers, RedPajama dataset	Continued pre-training on plant science corpus	Fine-tuned on 1030 instructions	Text → question answers
Chains-BERT ([112])	2023	BERT	Agricultural information database, Shandong farm management dataset	Transfer learning, training on unlabeled data	Contrastive learning based on text matching, semi-supervised learning	Text → classification labels, generated text
SAM ([113])	2023	ViT	Cage-free chicken dataset, broiler chicken dataset	-	-	Images → segmentation results, bounding box info (tracking)
SAM ([114])	2023	ViT	Leaf counting and segmentation challenge dataset	-	-	Image → segmentation masks
WDLm ([115])	2024	Transformer-based VLM	Wheat disease dataset, CGIAR dataset	Utilized pre-trained VLM results	LoRA fine-tuning, fine-tuned on disease dataset	Image, text prompts → disease classification, treatment suggestions

Building on the discussion of various AFMs, we now explore specific applications of FMs in agriculture, examining how they contribute to solving real-world agricultural challenges. These applications span several key areas, including Agricultural Knowledge Question Answering (AKQA), agricultural image and video analysis, agricultural decision-making, and agricultural robots and automation. Through case studies, we will explore the transformative potential of FMs in enhancing decision-making processes, improving crop management practices [116], advancing pest and disease detection, and driving the automation of farming tasks. By integrating large-scale data processing, NLP, and advanced CV techniques, FMs are playing a pivotal role in revolutionizing agricultural practices, rendering them more efficient and sustainable. Each of the following subsections will highlight the contributions, challenges, and future prospects of FMs in these domains.

(1) Agricultural Knowledge Question Answering: Systems leverage large-scale agricultural data and NLP technologies to provide automated, accurate responses to agriculture-

related queries. These systems integrate domain expertise to offer scientific support and decision-making guidance for farmers, researchers, and agricultural workers. Core functionalities include information retrieval, which extracts relevant answers from agricultural literature and expert knowledge, and automatic answer generation, which uses NLP models to deliver concise and precise responses. AKQA systems have significantly improved access to agricultural information, particularly in resource-scarce regions, reducing reliance on experts and enhancing daily farming practices. Applications span crop cultivation, pest management, and soil management, providing farmers with actionable insights.

Foundation language models have demonstrated remarkable success in AKQA. For example, large language models have demonstrated significant potential in agricultural applications, with performance varying substantially based on model architecture and augmentation techniques. In evaluations of the Certified Crop Advisor (CCA) certification exams, GPT-4 achieved a 93% accuracy rate when enhanced with RAG, representing a 14-point improvement over its baseline performance of 79% without retrieval augmentation. The comparative analysis revealed consistent performance hierarchies across model types, with GPT-4 outperforming GPT-3.5 (88% with RAG, 64% without) and substantially surpassing smaller models like Llama2-70B (81% with RAG, 55% without) and Llama2-13B (70% with RAG, 47% without), demonstrating how model scale and knowledge retrieval techniques collectively enhance agricultural question-answering capabilities. These results underscore the transformative potential of large language models in agricultural education and decision support, particularly when combining advanced architectures with domain-specific knowledge augmentation. Similarly, ChatAgri, based on ChatGPT, excels in agricultural text classification, particularly in crop cultivation and pest management tasks, outperforming traditional fine-tuning methods [110]. Chains-BERT combines BERT with semi-supervised and contrastive learning, enabling efficient question answering with minimal labeled data, making it highly adaptable to agricultural scenarios. The model achieves this through an innovative dual-encoder architecture, featuring a fixed feature extractor and trainable model that employs contrastive learning to optimize sentence representations. By augmenting this framework with Bi-LSTM layers for label-chain modeling and incorporating semi-supervised techniques like proxy labeling and consistency regularization, Chains-BERT effectively leverages both labeled and unlabeled agricultural data. Experimental results demonstrate its superior performance, achieving an 86.5 Micro-F1 score on the CAIL2018 dataset—a 5.5-point improvement over vanilla BERT. The model also exhibits strong generalization capabilities, outperforming baseline BERT by 3.25 F1 points on SQuAD 2.0, validating its effectiveness beyond agricultural domains [112]. In China, models like 'Hou Ji' and 'Shennong Model 2.0' have advanced AKQA by integrating IoT and multimodal reasoning, supporting diverse agricultural domains [111,117]. Additionally, VQA methods are emerging in crop disease diagnosis, combining image and text data for decision-making [118]. These advancements highlight the transformative potential of foundation models in AKQA, bridging gaps in agricultural knowledge dissemination.

(2) Agricultural Image and Video Analysis: Powered by CV and DL, this analysis plays a crucial role in crop health monitoring, pest detection, and yield prediction. Visual Foundation Models (VFM) process image and video data to provide accurate, real-time insights, enhancing agricultural intelligence. Traditional methods, such as molecular biology, are often time-consuming and invasive, whereas image-based approaches improve efficiency and reduce crop damage. For example, preprocessing techniques for night vision images have enhanced accuracy in apple harvesting robots [119]. Recent advancements include multispectral and hyperspectral imaging (MSI/HSI), which enable real-time monitoring of plant health and early disease detection [120–122].

VFMs are increasingly replacing traditional CNN models in crop disease diagnosis. The Wheat Disease Language Model (WDLN), for example, combines SAM and reasoning chains to isolate disease features in complex field environments, offering precise treatment recommendations [115]. Remote sensing data from drones and satellites further extend the application of VFMs, enabling large-scale crop health monitoring and pest identification [123,124]. VQA methods, such as multimodal feature fusion for fruit tree diseases, integrate image and text data to provide diagnostic support in areas lacking expert resources [118]. These technologies highlight the growing potential of VFMs in improving agricultural efficiency and sustainability.

(3) Agricultural Decision-Making: This entails complex decisions, such as crop selection, planting schedules [125], irrigation, and pest control, which directly impact productivity and sustainability. FMs enhance decision-making processes by providing data-driven insights and predictions [126]. For example, deep neural networks and GANs analyze historical yield data, weather patterns, and soil conditions to predict crop yields accurately [127,128]. These predictions enable farmers to optimize resource use and maximize productivity.

In pest and disease control, FMs analyze crop images to detect issues early, even during latent stages, reducing pesticide use and improving sustainability [129,130]. Intelligent systems like ‘The Farmer Chatbot’ provide real-time advice on plant protection, weather, and soil conditions, leveraging large datasets for accurate recommendations [131]. In livestock farming, FMs classify disease-causing pathogens and monitor animal health, improving resource allocation and welfare [132,133]. Additionally, FMs address climate change challenges by analyzing weather data and recommending adaptive strategies such as crop variety selection and water management [134]. Supply chain optimization is another key application, where FMs forecast demand and streamline production processes [135,136]. Despite their potential, challenges remain, including the need for real-time processing and user-friendly interfaces, which future research must address.

(4) Agricultural Robots and Automation: These technologies significantly enhance productivity by replacing manual labor with intelligent systems capable of crop monitoring, pest control, and harvesting. These robots integrate CV, DL, and sensing technologies to perform tasks such as precision fertilization and disease detection [137,138]. However, traditional systems face limitations in real-time data processing, prompting the adoption of FMs to improve scalability and efficiency.

FMs enable advanced functionalities in agricultural robots. For example, drone systems equipped with FMs monitor crop growth and predict yields in real time, while ground robots perform complex tasks like harvesting and sorting [139]. The SAM processes thermal images for crop segmentation and yield prediction, while the TAM enhances long-term disease monitoring [12,140]. Despite their advantages, FMs require significant computational resources, limiting their applicability in real-time scenarios [141]. Future research should focus on optimizing hardware and reducing resource consumption to fully realize the potential of FMs in agricultural automation.

4. Challenges of AFMs

This section discusses the challenges associated with the AFMs. It explores the complexity and heterogeneity of agricultural data, the difficulties in data acquisition, and the phenomenon of data shift. Additionally, it examines the practical implementation challenges faced when applying AFMs in real-world agricultural settings. Furthermore, this section underscores the necessity for advanced techniques to foster the advancement of AFMs. It provides insights into these challenges and potential solutions that will facilitate the effective application and expansion of FMs within the agricultural sector.

4.1. Diversity and Heterogeneity of Agricultural Data

Agricultural data are characterized by high diversity and heterogeneity [142], which is reflected not only in the types and sources of the data but also in their multifaceted impact on agricultural production, management decisions, and research [143]. With the rapid advancement of emerging technologies, such as precision agriculture and smart agriculture, agricultural data play an increasingly pivotal role in enhancing agricultural productivity, optimizing resource allocation, and mitigating environmental burdens. However, the complexity of agricultural data, coupled with their diversity and heterogeneity, renders them particularly challenging to utilize effectively in AFMs.

Firstly, the diversity of agricultural data sources constitutes a key factor contributing to data heterogeneity. Data can be collected through various channels, including weather stations, satellite remote sensing [144], drones, and IoT sensors. The formats, accuracy, and update frequencies of data from different sources vary, necessitating that FMs handle and integrate information from multiple data sources. For instance, climate data from weather stations and satellite remote sensing not only include indicators such as temperature and humidity but may also encompass additional factors, including wind speed and precipitation, which influence agricultural production. Soil data are typically collected through field testing or sensor monitoring, encompassing indicators such as pH levels, humidity, and organic matter content. Remote sensing data, obtained through satellite [145] or drone imagery, offer novel insights into agricultural practices.

Secondly, the application domains of agricultural data further illustrate their heterogeneity. Various types of data are often closely associated with specific tasks in agricultural production [146]. For example, climate data are essential for crop growth prediction, agricultural climate modeling, and disaster warning, whereas soil data play a pivotal role in soil management and crop planting decisions [147]. Remote sensing data facilitate crop monitoring, agricultural pest detection, and land use analysis, while water resource data are directly linked to irrigation management and water resource optimization [148]. Through precise crop growth monitoring and pest prediction [149], farmers can make more scientifically informed decisions regarding fertilization and irrigation. Each type of data possesses unique importance and distinct application scenarios.

Table 4 presents various types of data, their sources, and applicable fields in agriculture, thereby showcasing the diversity and heterogeneity inherent in agricultural data. Each data type can influence the outcomes of downstream tasks and may directly determine the efficacy of agricultural decisions. Consequently, effectively integrating these heterogeneous data sources and enabling FMs to comprehend and utilize this diverse information represents a significant challenge currently confronting AFMs.

Table 4. Overview of agricultural data sources and types.

Data Source	Data Type	Collection Method	Time Frequency	Advantages	Challenges
Satellite remote sensing	Climate, crop growth, etc.	Using satellite imagery for remote sensing	Daily/seasonal	Wide coverage, large-scale data acquisition	Resolution limitations, cloud interference
Drone imagery	Crop health, pest/disease	Drone flight for image capture	Daily/seasonal	High resolution, flexible operation	High cost, weather limitations
Ground sensors	Soil moisture, temperature, etc.	Installing sensors to monitor soil data	Real-time	Real-time data, high accuracy	High deployment and maintenance costs
Farmer reports	Crop growth status	Regular farmer reports on crop conditions	Weekly/seasonal	Large data volume, easy to schedule	Data accuracy issues, subjective bias
Agricultural research institutions	Specialized weather, crop growth	Based on professional models or field surveys	Seasonal	High professionalism, high-quality data	Limited data, small coverage range

Although the prospects for employing FMs in agriculture are promising, realizing this goal presents considerable challenges. The complexity and diversity of data necessitate

the development of algorithms capable of managing heterogeneous data, alongside substantial amounts of labeled data and efficient data fusion methods to enhance the model's generalization capabilities. Extracting meaningful information from multidimensional, multi-source data, eliminating noise, and constructing accurate agricultural models remain pressing challenges for researchers and practitioners.

4.2. Agricultural Data Acquisition

The challenges associated with acquiring agricultural data primarily arise from the complexity and decentralization of agriculture itself, particularly in developing countries and certain major agricultural nations where agricultural activities continue to rely on traditional, low-tech methods. The pervasive reliance on these traditional methods has resulted in the decentralization of agricultural production, and management systems have yet to achieve full automation. Consequently, data collection becomes particularly challenging. Agricultural production encompasses numerous factors, including soil, climate, water, fertilizers, crop cultivation, sunlight, and others. Each of these data points may exhibit significant variability due to regional differences, crop types, and climatic changes. Furthermore, the collection of these data faces dual challenges related to technology and funding, particularly in remote areas where data collection often necessitates substantial manual labor. Given the diversity and timeliness of agricultural data, the process is often time-consuming and labor-intensive.

Even in advanced agricultural nations, the standardization, unification, and accuracy of data collection continue to pose significant challenges. For instance, models for agricultural pests and diseases are frequently constrained by data quality and availability [150], and existing climate data regarding crop damage prediction face the challenge of becoming outdated [151,152]. To address this challenge, technological innovations, such as the introduction of Unmanned Aerial Vehicles (UAVs) [153] and IoT devices [154], alongside data collection methods like remote sensing and crowdsourcing [155], help mitigate some difficulties associated with traditional data collection. Nonetheless, challenges related to data privacy and ownership remain unresolved [156,157].

In this context, emerging technologies, such as GANs and self-supervised learning, have opened new avenues for agricultural data acquisition and utilization. Specifically, GANs and self-supervised learning techniques have demonstrated promising potential in enhancing data labeling efficiency and data generation. These techniques can simulate and generate data that are challenging to obtain in real-world scenarios, thereby providing additional data support for training agricultural models. These emerging technologies not only improve the efficiency of data acquisition but may also partially address issues such as data labeling and data scarcity [158,159].

Table 5 summarizes different sources of agricultural data acquisition and their associated challenges. It lists various data sources (such as satellite remote sensing, drone imagery, ground sensors, etc.), the types of data, acquisition methods, time frequencies, advantages, and challenges. By summarizing these data acquisition methods, we can better understand the strengths and weaknesses of each and provide more rational options and decision support for data collection in the agricultural sector.

Overall, the acquisition of agricultural data faces multifaceted challenges. These challenges stem not only from the complexity of agriculture itself but also from technological and resource limitations. While existing technologies have alleviated these issues to some extent, achieving comprehensive and efficient data collection still requires overcoming legal and ethical issues such as data privacy and ownership while promoting technological innovation and openness in data sharing.

Table 5. Overview of agricultural data types [160].

Data Type	Description	Possible Sources/Collection Methods	Application Fields
Climate data	Include temperature, humidity, precipitation, wind speed, etc., used to monitor the impact of climate change on agriculture.	Meteorological stations, satellite remote sensing, drones, meteorological APIs	Crop growth prediction, agricultural climate models, disaster early warning
Soil data	Include soil pH, moisture, fertility, organic matter content, etc., used to assess soil quality.	Soil testing, IoT sensors, remote sensing technologies	Soil management, crop planting decisions
Crop data	Include crop variety, growth stages, yield, pest/disease conditions, etc.	Ground observation, drone monitoring, satellite imagery, IoT sensors	Crop growth monitoring, pest prediction, fertilizer decision-making
Remote sensing data	Information obtained from satellite or drone images and videos.	Satellite remote sensing, UAVs, autonomous aerial vehicles	Crop monitoring, agricultural disease detection, land use analysis
Water resource data	Include river flow, groundwater levels, irrigation water usage, etc.	Hydrological stations, groundwater monitoring, remote sensing data	Irrigation management, water resource optimization, agricultural water analysis
Agricultural machinery data	Include machinery operational status, efficiency, maintenance records, etc.	IoT devices, GPS, sensors, agricultural robots	Precision farming, machinery scheduling, operational efficiency optimization
Agricultural production logs	Daily farm production activity records, such as sowing, fertilizing, spraying, etc.	Farm management systems, farmer manual records, sensor data	Agricultural management, crop production monitoring, agricultural decision support
Market price data	Data on the price fluctuations of different crops in the market.	Market surveys, E-commerce platforms, agricultural wholesale markets	Market analysis, price prediction, sales decisions
Agricultural environmental data	Environmental factors affecting crop growth, such as light, wind speed, and rainfall.	Environmental monitoring devices, meteorological data	Crop growth optimization, agricultural environmental analysis
Biodiversity data	Data on the population numbers and species distribution in agricultural fields.	Species surveys, remote sensing data, field observations	Ecological protection, crop–ecosystem interaction research

4.3. Agricultural Data Shift

Agricultural data shift refers to the significant discrepancies in data encountered during the training and deployment stages of models in agricultural applications. These shifts may arise from various factors, including environmental changes across different regions and seasons, differences in crop types, varying soil conditions, and diverse agricultural practices. Due to these environmental variations, the distribution of agricultural data can change substantially, impacting the performance and generalization ability of models. Table 6 summarizes the three main types of agricultural data shifts.

Table 6. Types of agricultural data shift and their manifestations.

Shift Type	Definition	Manifestation in Agriculture	Example
Concept shift	The distribution of target variables changes over time or due to environmental changes.	Changes in crop growth cycles, yield, or other agricultural factors due to climate change or production shifts.	Certain crops may mature earlier or later due to weather changes, leading to mismatches in data features and labels [161].
Covariate shift	The distribution of feature variables changes, but the label remains the same.	Geographic and seasonal variations in soil types, climate conditions, and crop varieties.	A crop classification model trained in one region may perform poorly in another region due to differences in crop features such as leaf color and texture [162,163].
Label shift	The distribution of labels changes.	Crop yield and quality may fluctuate due to climate and environmental changes, impacting label stability.	A crop yield might decrease significantly in a given year due to extreme weather, affecting the label's stability [164].

Label shift pertains to changes in the distribution of labels across different time periods or environments. For instance, crop yields are influenced by climate and environmental factors, resulting in fluctuations in yield labels for crops across various years or regions. Some crops may experience yield reductions or even fail to grow due to extreme weather conditions, thereby affecting the stability of labels [164].

Data shifts in agriculture have been confirmed by multiple studies. For example, [165] identified a high correlation between certain land cover categories, such as medium-density residential areas and dense residential areas, or between buildings and water storage tanks. This correlation can lead to confusion in classification models, adversely impacting accuracy. Similarly, [164] observed classification confusion between corn and soybeans, while [163] noted significant confusion among tree crops, summer crops, and vegetable cultivation. In the context of plant phenotype recognition, [166,167] highlighted challenges in distinguishing between different plant phenotypes during growth stage transitions, as the appearance of plants changes gradually. This includes not only the slow transition of plant surface features but also variations in climate, soil types, and field management practices, leading to significant differences in data feature distribution over time. Regarding fruit counting, [168,169] found that occlusion, height differences, and unstable lighting conditions are primary factors affecting accuracy. The similarity in color between fruits and leaves exacerbates this issue, diminishing model performance and suggesting that environmental changes can induce shifts in both feature and label data, thereby affecting the generalization ability of models. Furthermore, studies by [170,171] and [172] have examined the differentiation between weeds and crops, particularly concerning features such as shape, texture, color, and position. Due to the similarities between weeds and crops, especially when weeds obscure crops, models often struggle to classify them accurately. This situation is closely related to data distribution shifts, as the same weed may present different features in various environments, impacting classification performance.

Research indicates that data shifts in agriculture can significantly affect model performance, potentially leading to a sharp decline in effectiveness. For example, [114] demonstrated that applying FMs to zero-shot leaf segmentation tasks resulted in poor model performance, likely due to data distribution shifts. To address these challenges, researchers have proposed several solutions, including multi-task learning, continual learning, and distillation techniques to mitigate performance declines caused by data shifts [173].

To effectively tackle agricultural data shifts, scholars have suggested various strategies and methods. These include multi-task learning and continual learning, which enable models to adapt to new data distributions during training, thereby enhancing their generalization ability [173]; domain adaptation techniques, where domain-adaptive models can transfer learned knowledge from training data to different environments, reducing the impact of data shifts [174]; and data augmentation and mixing techniques, which generate diverse training data, particularly in terms of climate, soil, and crop types, effectively bolstering the robustness of models to mitigate data shifts.

4.4. Time Lag Issues

The time lag issue in agricultural data presents a significant challenge. Crop growth cycles influenced by seasonal and climatic variations cause training data for AFMs to gradually lose relevance [2]. Post-deployment, time-series data may become outdated due to changing environmental conditions, leading to prediction deviations [175]. In agricultural vehicular networks, transmission delays between mobile units can compromise decision-making with stale data, particularly affecting real-time applications like GPS navigation. For agricultural robots performing time-sensitive tasks (e.g., harvesting, pest

control), processing delays > 200 ms directly impact accuracy and yield, exacerbated by foundation models' computational demands [176].

Delay-Tolerant Networks (DTNs) address transmission delays in remote areas, as demonstrated by PotatoScanner's store-carry-forward approach [177]. For robotics, DTN-edge computing hybrids maintain real-time responsiveness in large fields, while model pruning/quantization reduces inference latency for harvesting operations [178].

Time series analysis and dynamic modeling help accommodate agricultural data's temporal dependencies [179]. Lightweight Transformer architectures now balance accuracy and efficiency for real-time field automation [180], while continual learning maintains model relevance.

4.5. Practical Deployment Challenges

The operational deployment of AFMs faces fundamental technical constraints that manifest differently across farm scales. While current automation systems like Kenya's FarmBot demonstrate basic inclusivity for smallholders through compressed vision models [181], the integration of full-fledged FMs in low-connectivity environments remains constrained by substantial computational demands. Recent breakthroughs in extreme model compression show promise—binary transformers achieve 99% size reduction while retaining 80% original accuracy in controlled settings, potentially enabling localized FM deployment [182]. However, even distilled models confront memory limitations, as ViT-base architectures require >1.7GB memory, while edge devices typically offer <512 MB RAM, forcing difficult tradeoffs between functionality and accessibility [130].

In real-world operating conditions, these challenges become more acute. Agricultural data streams exhibit inherent asynchrony (e.g., 15–300 ms latency between visual and sensor inputs), which significantly degrades decision accuracy without temporal alignment modules—a particular concern for transformer-based multimodal agricultural foundation models where self-attention mechanisms are temporally sensitive. For smallholder farmers, this complexity is addressed through accessible “Model-as-a-Service” platforms rather than local deployment. Farmers can simply upload field photos via mobile apps to access cloud-based FM APIs. These solutions feature carefully designed interfaces with voice input and visual output to build trust in AI recommendations while employing progressive deployment strategies, starting with basic tasks before advancing to complex decision support. This approach democratizes access to advanced FM capabilities without requiring expensive infrastructure, though it introduces 40–60% higher cloud dependency costs compared to industrial on-premise solutions [183].

5. Development Directions for AFMs

The ongoing advancement and deployment of AFMs signify a transformative phase for the agricultural sector. Future iterations of these models are expected to exhibit enhanced capabilities that extend beyond conventional text and image processing to encompass comprehensive multimodal integration. By leveraging their inherent multimodal architecture, next-generation AFMs will effectively synthesize heterogeneous data streams, including video feeds, audio recordings, and sensor outputs. This technological evolution promises to catalyze unprecedented levels of intelligent automation across agricultural systems, facilitating more sophisticated decision-making frameworks and operational optimization.

To fully grasp the potential of AFMs, it is essential to explore their development from multiple perspectives. The following sections will delve into the key areas where AFMs are anticipated to make significant impacts: leveraging multimodal data, integrating AFMs across the agricultural and food sectors, enhancing intelligent decision-making systems, and addressing ongoing technical development and talent training challenges. Each of

these aspects will be discussed in detail to provide a comprehensive understanding of how AFMs are poised to revolutionize agriculture.

5.1. Leveraging Multimodal FMs in Agriculture

Future AFMs should expand beyond traditional text and image modalities to harness emerging technological capabilities. This multimodal expansion forms the technical foundation for cross-domain integration across agricultural supply chains. Recent studies demonstrate the untapped potential of multimodal integration in agriculture: video analytics enable real-time crop monitoring during critical growth stages [184,185], while audio signal processing shows promise in non-destructive quality assessment [186,187].

The intrinsic value of multimodal data lies in their ability to bridge discrete phases of the agricultural value chain. For instance, combining harvest-stage audio ripeness data with growth-period visual analytics allows AFMs to predict post-harvest storage requirements and optimal market timing [188]. Such temporal–spatial data fusion enables vertically integrated decision-making, from field management to logistics optimization.

This technological convergence directly addresses supply chain fragmentation challenges. By fusing complementary data streams—from spectral signatures to temporal growth patterns—multimodal AFMs can synchronize production data with downstream processing parameters (e.g., matching harvest quality metrics to storage facility conditions). The resulting closed-loop intelligence system not only optimizes field-level production efficiency but also enhances the resilience of agricultural supply networks through data-driven coordination [189].

5.2. Integrating AFMs Across the Agricultural and Food Sectors

Agriculture is intrinsically linked to the entire food production and supply chain [190], and the development of AFMs is poised to drive technological advancements across various stages of this chain. Beyond the food industry, AFMs will play a critical role in optimizing agricultural production, improving resource management, and enhancing decision-making processes from crop planting to food processing and distribution.

In the agricultural production phase, AFMs can integrate data from diverse sources, such as soil sensors, weather stations, and crop health monitoring systems, providing real-time insights and predictive analytics for farmers. Recent research demonstrates that multimodal fusion techniques are critical for handling heterogeneous agricultural data. For instance, the Rice-Fusion framework combines CNN-processed leaf images ($200 \times 200 \times 3$ RGB) with MLP-analyzed agro-meteorological data (temperature, humidity, NPK values) through feature-level concatenation. This early fusion approach aligns temporal sensor readings (minute-level) with daily drone images via timestamp synchronization, achieving 95.31% disease diagnosis accuracy—a 12.8% improvement over unimodal models. The framework further addresses modality disparities by normalizing sensor data (min–max scaling) and image pixels (0–1 range) before fusion, ensuring compatibility between numerical and visual features [191]. These models can guide farmers in making informed decisions regarding irrigation, pest control, and fertilization, ultimately improving crop yields and minimizing resource waste. Furthermore, AFMs can assist in monitoring the effects of climate change on crop growth, enabling proactive strategies to adapt to shifting environmental conditions.

In the food processing sector, AFMs can facilitate quality control and traceability [192,193], enabling real-time tracking of food from farm to table. By incorporating AFMs into food safety monitoring systems, producers can swiftly identify potential contamination risks and ensure compliance with food safety standards. This seamless integration across agricultural

and food production stages will enhance the transparency of food supply chains, which is crucial for fostering consumer trust [194].

Moreover, AFMs can significantly improve supply chain management by predicting demand fluctuations [195], optimizing logistics, and reducing waste. For instance, predictive models could forecast crop availability and help align supply with market demand, leading to more efficient food distribution and minimizing food loss.

In the future, the integration of AFMs with other models across different sectors in the agricultural and food industries will create a mutually reinforcing ecosystem. This cross-sector collaboration will elevate the intelligence of agriculture, food production, logistics, and quality assurance systems, ultimately ensuring the sustainability, safety, and efficiency of the entire food supply chain [184,196].

5.3. Intelligent Decision-Making Systems Based on AFMs

AFMs possess immense potential for enhancing intelligent decision-making, particularly in areas such as crop planting, pest management, irrigation, and fertilization [197]. By analyzing multi-source data, AFMs can facilitate intelligent decision-making and predictions, enabling farmers to make more precise management choices. For instance, intelligent planting and breeding decision systems can integrate sensor data, environmental information, and agricultural expertise to automatically provide farmers with scientifically grounded recommendations [196]. As FMs are increasingly applied across various agricultural domains, the future of agricultural production is set to advance toward greater precision, automation, and efficiency.

Future development of AFM-based decision systems will follow a phased technical roadmap, with an initial focus on lightweight mobile applications delivering basic agricultural guidance through model compression, intermediate development of distributed learning architectures for cross-regional knowledge sharing and cost optimization, and ultimately evolving into self-adaptive intelligent decision systems. Implementation will incorporate data governance protocols, tiered service models, and farmer-centric interface design to ensure equitable access across farm scales, progressively bridging the digital divide in agricultural intelligence adoption.

6. Conclusions

This paper provides a comprehensive overview of the development, applications, and challenges of FMs in agriculture, with a particular focus on AFMs. Through an in-depth exploration of the history, key technologies, and architectures of FMs, this work highlights the transformative potential of these models in revolutionizing agricultural practices. The paper also offers valuable insights into the process of building FMs, encompassing data selection, model architecture design, training and optimization, and evaluation metrics.

In addition to examining the fundamental aspects of FMs, this paper delves into their specific applications in agriculture, including agricultural knowledge question-answering, image and video analysis [198], decision support systems, and agricultural robotics and automation. It further explores the unique challenges posed by the agricultural domain, such as data diversity, acquisition issues, data shifts, time lags, and concerns regarding data privacy and ownership. Through this detailed analysis, this paper elucidates how AFMs can address these challenges, ultimately advancing the digitalization and intelligent transformation of the agricultural sector.

The development directions for AFMs have been outlined in this paper, emphasizing the importance of leveraging multimodal FMs to enhance decision-making processes in agriculture. The integration of AFMs across the agricultural and food sectors promises to optimize resource management and food supply chain systems, while the advancement

of intelligent decision-making systems will further improve the precision, efficiency, and automation of agricultural production. Additionally, ongoing technical developments and talent training are identified as critical factors for the future success and widespread adoption of AFMs in agriculture.

By summarizing the state-of-the-art research and providing a clear roadmap for future directions, this paper aims to contribute to the further development of agricultural intelligence, helping to propel the sector toward more sustainable and efficient practices. The application of AFMs, coupled with continuous advancements in technology and expertise, will play a crucial role in ensuring global food security and achieving ecological balance in agricultural production.

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